In this chapter we will introduce a new deep learning architecture that goes some way to unraveling this problem. Moreover, it not only offers interpretability gains but also performance gains in many cases. This new class of models is called *attention models* because they learn how to *attend to* (or focus on) only the salient aspects of an input. More specifically for our case, we will be developing a *self-attention model*, which is a model that allows each feature within an input to learn to attend to various other features in the input. This form of attention is closely related to the class of neural networks termed *graph neural networks*, which are neural networks explicitly designed to operate on graph structured data.

10.1. Machine learning interpretability with attention and relational biases

A *graph* (also called a network) is a data structure that is composed of a set of *nodes* and *edges* (connections) between nodes (figure 10.1). The nodes could represent anything: people in a social network, publications in a publication citation network, cities connected by highways, or even images where each pixel is a node and adjacent pixels are connected by edges. A graph is a very generic structure for representing data with relational structure, which is almost all the data we see in practice. Whereas *convolutional neural networks* are designed to process grid-like data, such as images, and *recurrent neural networks* are well-poised for sequential data, graph neural networks are more generic in that they can handle any data that can be represented as a graph. Graph neural networks have opened up a whole new set of possibilities for machine learning and they are an active area of research (figure 10.2).

Figure 10.1. A simple graph. Graphs are composed of nodes (the numberlabeled circles) and edges (the lines) between nodes that represent relationships between nodes. Some data is naturally represented with this kind of graph structure, and traditional neural network architectures are unable to process this kind of data. Graph neural networks (GNNs), on the other hand, can directly operate on graph-structured data.